

White paper: the three carryover analysis specifications, and why these three

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pmsimstats team. A design-justification note for the carryover-sensitivity manuscript (`analysis/report/02-carryover-sensitivity/report.Rmd`). It names the three analysis-model specifications the study compares, states them in the canonical notation of `analysis/report/NOTATION.md`, and sets out why these three, rather than some other three, are the right set to evaluate. It closes with the one specification the current set omits, and with three gaps in the canonical notation that the comparison brings to light.

Intended reader. A statistically literate physician: someone comfortable with regression and with the idea of an interaction term, but who should not be expected to supply the simulation-methods background. Statistical terms are glossed at first use.

1. The clinical problem, and the choice it forces

In an aggregated N-of-1 trial, each patient is crossed repeatedly between active drug and comparator, and many such single-patient series are pooled into one mixed model. The question these trials exist to answer is whether a baseline measurement predicts who benefits, that is, whether a **predictive biomarker** identifies the patients in whom the drug works. Statistically this is a biomarker-by-treatment **interaction**: the treatment effect is allowed to differ according to the patient's baseline biomarker value.

The complication is that drugs do not stop working the moment they are stopped. The effect decays over days to weeks, so measurements taken during a nominally off-drug period still reflect partial drug exposure. This residual effect is called **carryover**. In the prazosin and post-traumatic stress disorder setting this program is calibrated to, the relevant half-life is on the order of half a week to a week, while measurements are weekly, so the first off-drug measurement typically still carries roughly half the drug effect.

This forces a decision on the analyst that has no default answer. The model needs a variable representing drug exposure at each occasion. Coding that variable as simply on or off is one choice among several, and the manuscript exists to determine which choice performs best. The present note explains why the field of candidates was narrowed to three.

1.1 The three trial designs

The comparison is run under three protocols, which differ in how the on-drug and off-drug measurement occasions are arranged. Because the whole problem concerns what happens *after* the drug stops, a design that generates more post-discontinuation observations gives the analyst more carryover to handle, and the ranking of the three specifications turns out to depend on which design is used. All three use eight measurement occasions per patient. Patients are randomized between two or four **paths**, meaning fixed schedules of on-drug and off-drug periods.

- **CO (crossover).** The traditional two-period within-patient crossover. One path takes drug for the first four visits and stops; the other takes comparator first and starts drug at visit five. Half the patients therefore never discontinue at all.
- **OL+BDC (open-label plus blinded discontinuation).** Patients are titrated openly on active drug, then blindly switched to placebo. Both paths discontinue, but late, so most visits are on drug and only two or three follow discontinuation.
- **Hybrid.** The Hendrickson design and this program's reference: open-label titration, then blinded discontinuation, then a brief crossover at the end. Four paths, and the richest in on-off transitions, which is why it is used as the reference design throughout.

2. Notation

Symbols follow `analysis/report/NOTATION.md`, the compendium's single source of truth for notation. The plain-language column is added here for readability and is not part of that file.

Symbol	NOTATION.md meaning	In plain terms
Y_{it}	observed symptom score for patient i at timepoint t (code column Sx)	the outcome being measured
B_i	pre-treatment biomarker value (code bm)	the baseline test result
b_i	standardized biomarker, $(B_i - \bar{B})/s_B$	the same, rescaled so one unit is one standard deviation
D_{it}	binary drug state: 1 on drug, 0 off drug	the simple on/off switch
$D_{bc,it}$	continuous exposure-decayed indicator: 1 on drug, $e^{-\lambda t_{sd}}$ off drug (code Dbc)	a dimmer switch that fades out after the drug stops
t_{sd}	time since discontinuation (code tsd)	weeks since the drug was stopped
$t_{1/2}$	carryover half-life (code carryover_t1half)	time for the residual effect to halve
λ	decay rate, $\ln 2/t_{1/2}$	how fast the residual effect fades
u_i	patient random intercept	each patient's own baseline level
ε_{it}	observation-level residual	unexplained measurement-to-measurement variation
c_{bm}	moderation parameter of the covariance-moderation architecture	how strongly the biomarker predicts benefit
σ_{BR}	standard deviation of the response component	the scale the effect is expressed on
θ_{true}	calibrated true value, $-c_{bm}\sigma_{BR}$	the correct answer the simulation is checked against
N	total patients, across all randomization paths	trial size

A hat marks an analyst-side quantity as against a true one, so $\hat{t}_{1/2}$ is the half-life the analyst assumes while $t_{1/2}$ is the one the data were generated with. Per rule 5 of **NOTATION.md**, the response components are non-negative *reductions* in symptom severity, so a beneficial treatment effect lowers Y_{it} and interaction coefficients are negative.

Three symbols this note requires are not in **NOTATION.md**. They are used here provisionally and proposed formally in Section 9: L_{it} for the lagged just-off-drug indicator, $\beta_{bm:D}$ disambiguated from $\beta_{bm:D_{bc}}$ according to which exposure variable carries the interaction, and the main-effect coefficients β_b , β_t , β_D , β_L . Note that β_b , the biomarker main effect, is a different object from β_{bm} , which **NOTATION.md** reserves for the moderation parameter of the mean-moderation architecture.

Abbreviations used below, per the **NOTATION.md** glossary: **CO** (crossover), **BDC** (blinded discontinuation), **OL+BDC** (open-label titration followed by blinded discontinuation), **Hybrid** (the Hendrickson design: open-label, then blinded discontinuation, then a brief crossover), **LME** (linear mixed-effects model), **MCSE** (Monte Carlo standard error), and **ADEMP** (the Morris, White and Crowther reporting framework for simulation studies).

All three specifications share one model,

$$Y_{it} = \beta_0 + \beta_b b_i + \beta_t t + \beta_D X_{it} + \beta_{bm:X} b_i X_{it} + u_i + \varepsilon_{it},$$

fitted as an LME with a patient random intercept and **corCAR1** residual correlation, which allows measurements taken closer together in time to be more similar. In words: the outcome depends on the biomarker, on time, on drug exposure, and on the product of biomarker and exposure. That last product term, $\beta_{bm:X}$, is the estimand, meaning the specific quantity the trial is trying to estimate. It says how much the treatment effect changes per standard deviation of the biomarker.

The three specifications differ only in how the exposure variable X_{it} is built, and in whether an extra term is added.

3. The three specifications

3.1 Unadjusted

$$X_{it} = D_{it}, \quad D_{it} \in \{0, 1\}$$

In code, $Sx \sim bm + t + Db + bm:Db$. Carryover is not represented at all. A measurement taken one week after stopping, still carrying about half the drug effect, is coded exactly like one taken after the drug has fully cleared. The estimand is $\beta_{bm:D}$, the coefficient on $b_i D_{it}$.

This is what an analyst gets by default from a standard mixed model without thinking about carryover, which is precisely why it belongs in the comparison: it is the benchmark any remedy must beat to justify its extra complexity.

3.2 Lag-adjusted

$$X_{it} = D_{it}, \quad L_{it} = D_{i,t-1}(1 - D_{it})$$

with L_{it} added as an extra term, in code $Sx \sim bm + t + Db + bm:Db + L$. The indicator L_{it} equals 1 at the first off-drug measurement following an on-drug one, and 0 everywhere else. It flags the occasions most likely to be contaminated by residual drug and lets the model estimate how much those occasions differ, rather than requiring the analyst to say in advance how fast the drug washes out. This is the standard remedy in the crossover-trial literature, following Jones and Kenward.

Two structural features determine how its results must be read. First, L_{it} is added only as a main effect, with no $b_i L_{it}$ product term. It can therefore adjust the average outcome at contaminated occasions, but it cannot adjust how the *biomarker effect* behaves there. Second, it involves no half-life, so it is completely unaffected by what the analyst assumes about the pharmacokinetics. The estimand is $\beta_{bm:D}$, the same quantity Unadjusted targets.

3.3 Exposure-weighted

$$X_{it} = D_{bc,it} = \begin{cases} 1 & \text{if } D_{it} = 1, \\ \exp(-\hat{\lambda} t_{sd,it}) & \text{if } D_{it} = 0, \end{cases} \quad \hat{\lambda} = \frac{\ln 2}{\hat{t}_{1/2}}$$

In code, $Sx \sim bm + t + Dbc + bm:Dbc$. Instead of an on/off switch, exposure is a dimmer that starts at 1 and halves every $\hat{t}_{1/2}$ weeks after the drug stops. A measurement one week after stopping, at an assumed half-life of one week, enters as 0.5 rather than 0. This is the current default in the `pmsimstats-ng` package.

It is the only one of the three requiring the analyst to commit to both a decay shape and a half-life. Because those assumptions enter through the values of the variable rather than through the model formula, it is effectively a different model for every assumed half-life even though it is written identically each time. Its estimand is $\beta_{bm:D_{bc}}$, the coefficient on $b_i D_{bc,it}$, which is *not* the same quantity as $\beta_{bm:D}$.

3.4 The three codings on one patient

The differences are easiest to see on a single schedule. The table below takes one OL+BDC path and one Hybrid path, and shows what each specification writes into the model at each visit, assuming a half-life of one week. Verified by direct computation from the design definitions in `simulation-core.R`.

OL+BDC, path A. Drug is stopped after week 18.

Week	On drug	t_{sd}	Unadjusted D_{it}	Lag flag L_{it}	Exposure-weighted $D_{bc,it}$
4	yes	0	1	0	1
8	yes	0	1	0	1

Week	On drug	t_{sd}	Unadjusted D_{it}	Lag flag L_{it}	Exposure-weighted $D_{bc,it}$
12	yes	0	1	0	1
16	yes	0	1	0	1
17	yes	0	1	0	1
18	yes	0	1	0	1
19	no	1	0	1	0.50
20	no	2	0	0	0.25

Read the last two rows. At week 19 the patient still has about half the drug effect on board. Unadjusted records a 0, exactly as it does for a drug-free patient. Lag-adjusted records a 0 as well, but flags the visit so the model can allow its average outcome to differ. Exposure-weighted records 0.50, which is the value that actually reflects the residual exposure. At week 20 the difference persists: Unadjusted and Lag-adjusted both record 0 for a visit still carrying about a quarter of the effect, and only Exposure-weighted distinguishes it.

Hybrid, path A. Drug is stopped after week 10, restarted at week 16, and stopped again after week 16.

Week	On drug	t_{sd}	Unadjusted D_{it}	Lag flag L_{it}	Exposure-weighted $D_{bc,it}$
4	yes	0	1	0	1
8	yes	0	1	0	1
9	yes	0	1	0	1
10	yes	0	1	0	1
11	no	1	0	1	0.50
12	no	2	0	0	0.25
16	yes	0	1	0	1
20	no	4	0	1	0.0625

This path exposes the weakness of the lag flag directly. Weeks 11 and 20 both receive $L_{it} = 1$, because each is the first visit after a discontinuation, and the model therefore assigns them a single shared coefficient. But week 11 is one week after stopping and retains about 50% of the drug effect, while week 20 is four weeks after stopping and retains about 6%, an eightfold difference. Exposure-weighted separates them, 0.50 against 0.0625, because it is built from elapsed time, whereas the lag flag is built from position in the visit sequence and cannot tell the two apart. This is the mechanism behind the Hybrid caveat in Section 5.2.

4. Why these three

Four arguments. The second is the substantive one.

4.1 They span the three possible positions on washout

Write $\phi(t_{sd})$ for the fraction of drug effect still present t_{sd} weeks after stopping. Every analyst takes a position on ϕ , whether or not they state one, and only three positions are qualitatively distinct:

- **Deny it.** Assume $\phi \equiv 0$, so exposure is strictly on or off. This is Unadjusted.
- **Estimate it.** Leave the shape of ϕ unspecified but give it one free parameter, applied at the first off-drug occasion. This is Lag-adjusted.
- **Assume it.** Commit to a specific decay curve and rate, $\phi(t_{sd}) = \exp(-\hat{\lambda}t_{sd})$. This is Exposure-weighted.

The three are therefore not an arbitrary sample from a large space of codings. They are one representative of each available stance toward the unknown washout, ordered by how much the analyst is willing to assume.

There is also a formal relationship that makes this ladder continuous rather than merely rhetorical. As the assumed half-life goes to zero, the dimmer switch collapses to the on/off switch, since $t_{1/2} \rightarrow 0$ gives $\hat{\lambda} \rightarrow \infty$ and hence $D_{bc,it} \rightarrow D_{it}$. **Unadjusted is therefore the limiting case of Exposure-weighted at an assumed**

half-life of zero, not a separate method sitting outside it. At the other extreme, as $\hat{t}_{1/2} \rightarrow \infty$, the dimmer never dims, $D_{bc,it} \rightarrow 1$ everywhere, the on-off contrast disappears and the model can no longer estimate anything. The manuscript's half-life-mis-specification sweep is consequently a traverse along a path whose left endpoint is Unadjusted itself, and Unadjusted should be read as the anchor of that curve.

4.2 They separate the two ways carryover does damage

This is the strongest reason for this particular set, because it makes the comparison a controlled experiment rather than a horse race.

Carryover degrades the interaction test through two distinct mechanisms:

- **Dilution.** The exposure variable is a mis-measured version of the exposure the patient actually received. Contaminated occasions are entered as though they were drug-free, so the estimated interaction is pulled toward zero. This is the same phenomenon clinicians know as regression dilution, where a single noisy blood-pressure reading understates the true association with outcome. The signal shrinks.
- **Added noise.** Those same occasions differ from one another in ways the model does not describe, since some are one week off drug and others four. That unexplained variation is absorbed into the residual ε_{it} , which inflates the standard error of the interaction estimate. The noise grows.

Power falls on both counts at once, the signal shrinking while the noise grows. The three specifications occupy three different cells of the resulting classification:

Specification	Dilution	Added noise
Unadjusted	not addressed	not addressed
Lag-adjusted	not addressed	addressed at the first off-drug occasion
Exposure-weighted	addressed	addressed

The placements follow from where each term sits. Lag-adjusted adds L_{it} to the average-outcome part of the model only, so it can absorb noise from contaminated occasions, but because the interaction is still carried by the on/off variable D_{it} , the dilution is untouched. Exposure-weighted instead replaces the variable *inside* the interaction, so it addresses dilution directly and mops up the associated noise as a by-product.

Two things follow, and together they are the reason this set is well chosen:

1. **Comparing Lag-adjusted with Unadjusted isolates the noise mechanism.** The two estimate the same quantity, are fitted to the same simulated patients, and differ by exactly one term. Any difference between them is therefore attributable to the noise mechanism alone, with nothing else varying. Controlled contrasts this clean are rare in a simulation study.
2. **What is left over is interpretable.** Whatever advantage Exposure-weighted shows beyond that first comparison is the value of fixing dilution. The design therefore reveals not only which method wins but which of the two mechanisms actually matters in this setting, which is the more portable finding.

A set of three methods that all attacked the same mechanism, or that differed on several dimensions at once, would support neither conclusion.

4.3 Each is the working default of a literature

Each specification is the standard approach of a distinct research tradition bearing on this problem, so the comparison speaks to all three audiences rather than one:

- Unadjusted is ordinary applied practice in aggregated N-of-1 analysis, and the default of any off-the-shelf mixed model.
- Lag-adjusted is the crossover-trial tradition, where a lagged nuisance term is the orthodox remedy for carryover.
- Exposure-weighted is the pharmacokinetically motivated approach, used by Hendrickson and implemented as the pmsimstats-ng default.

Omitting any one would expose the study to the objection that it never tested the method its readers actually use.

4.4 They are equally estimable, so no cell is decided by failure

All three can be fitted in all three trial designs (CO, OL+BDC and Hybrid) at both sample sizes, need no data beyond what the trial already collects, and add at most one term to the model. Verified by direct recomputation from the stored cell-level summaries: across the full 540-cell production grid, none of the three produced a missing power value or a missing estimate. This matters because a comparison in which one method quietly fails to converge in some conditions is not a comparison of methods but of numerical robustness.

5. What the set excludes, and the one genuine gap

5.1 Candidates set aside

Other specifications were considered and not carried forward:

- **Washout deletion**, discarding off-drug measurements taken within some window of stopping. This is exposure weighting with crude 0/1 weights and an arbitrary cutoff, so it is a coarser version of Exposure-weighted, and it discards observations the other methods use.
- **Estimating the half-life from the data** rather than fixing $\hat{t}_{1/2}$ in advance. This is the principled answer to Exposure-weighted's main weakness and is the natural next step, but the half-life is poorly determined at $N = 35$ from a handful of off-drug measurements per patient, so it introduces a convergence-failure mode the other three do not have.
- **A full set of lag indicators**, one per off-drug occasion. The off-drug window here is only three to four measurements long, so a full set is nearly indistinguishable from the time term t and exhausts the available information.

5.2 The genuine gap: the $b_i L_{it}$ specification

The classification in Section 4.2 has an empty cell. No specification addresses dilution *within the lag family*. That specification would be

$$Y_{it} = \beta_0 + \beta_b b_i + \beta_t t + \beta_D D_{it} + \beta_L L_{it} + \beta_{bm:D} b_i D_{it} + \beta_{bm:L} b_i L_{it} + u_i + \varepsilon_{it},$$

in code `Sx ~ bm + t + Db + L + bm:Db + bm:L`. The current Lag-adjusted specification has β_L but not $\beta_{bm:L}$, and that one missing term is the entire difference.

Why the extra term reduces dilution. When the interaction is carried only by $b_i D_{it}$, the estimate $\beta_{bm:D}$ compares on-drug occasions against *all* off-drug occasions lumped together. Because that comparison group still contains contaminated occasions, which display part of the biomarker effect themselves, the contrast is blurred. Adding $b_i L_{it}$ takes the most contaminated occasions out of the comparison group and gives them their own coefficient. What is left behind is a cleaner comparison, so $\beta_{bm:D}$ recovers more of the true effect.

This makes the specification the assumption-free twin of Exposure-weighted. Exposure-weighted asserts that the biomarker effect at an off-drug occasion is the on-drug effect multiplied by $\exp(-\hat{\lambda} t_{sd})$, a fixed curve chosen in advance. The $b_i L_{it}$ specification instead estimates a free two-step pattern, full effect on drug and $\beta_{bm:L}$ at the first off-drug occasion, committing to no decay shape at all. Same goal, opposite stance on whether the washout is assumed or estimated.

Would it work in these designs? The table below reconstructs the exposure pattern of every randomization path, applying the same lag rule the pipeline uses, and reports how many weeks after stopping the flagged occasions sit, alongside those left in the comparison group. Verified by direct computation from the design definitions in `simulation-core.R`; t_{sd} is in weeks.

Design	Path	D_{it} pattern	Flagged ($L = 1$)	t_{sd} where $L = 1$	t_{sd} in comparison group
CO	A	1 1 1 1 0 0 0 0	1	1	2, 3, 4
CO	B	0 0 0 0 1 1 1 1	0	none	never on drug
OL+BDC	A	1 1 1 1 1 1 0 0	1	1	2
OL+BDC	B	1 1 1 1 1 0 0 0	1	1	2, 3
Hybrid	A	1 1 1 1 0 0 1 0	2	1, 4	2
Hybrid	B	1 1 1 1 0 0 0 1	1	1	2, 6
Hybrid	C	1 1 1 0 0 0 1 0	2	1, 4	2, 3
Hybrid	D	1 1 1 0 0 0 0 1	1	1	2, 3, 7

Three consequences follow, and each tempers the expected benefit.

- **The improvement would be partial.** The comparison group still contains occasions two or more weeks after stopping, which at a one-week half-life retain about 25% of the drug effect and less. The extra term removes the worst contamination and leaves the rest, so it should reduce dilution rather than eliminate it.
- **Half the crossover sample contributes nothing.** Path B of the CO design starts off drug, goes on, and never stops, so it contains no flagged occasions at all. The new coefficient would be estimated from path A alone.
- **The flag is crude in the Hybrid design.** Paths A and C each carry two flagged occasions, one at one week after stopping and one at four weeks, holding roughly 50% and 6% of the drug effect respectively. A single indicator pools them under one coefficient. This is exactly the case that favors using elapsed time directly, as Exposure-weighted does, because the flag is defined by position in the measurement sequence and is blind to the uneven spacing of the Hybrid visits.

Expected effect, marked as inference. This specification has not been run. Reasoning from the residual-exposure arithmetic above, it should move the estimate at the Hybrid reference cell from about -2.51 , where Unadjusted and Lag-adjusted both sit, partway toward the correct value $\theta_{\text{true}} = -3.6$ without reaching it, and should recover some power. Whether it closes the roughly ten-percentage-point gap to Exposure-weighted at that cell is genuinely open, and the third bullet argues it will not close it under the Hybrid design specifically.

Cost. The code change is small: one branch in `fit_spec()` and one more level in the grid, with no change to the data-generating side. A four-arm paper must be consistent across every table it prints, however, so it requires re-running the Tier 1 factorial (roughly 85 minutes on eight cores), the Tier 2 sensitivity blocks (about four minutes) and the S6 recalibration, then re-summarizing and regenerating all figures. The `quick-sim` rerun would not pick up the fourth arm automatically, since it runs through the package rather than `simulation-core.R`. The larger cost is editorial: the manuscript’s title, abstract and discussion are all organized around a two-way contrast, and a fourth arm means restructuring that spine a second time.

Recommendation: a staged approach rather than a blind fourth arm. Run $b_i L_{it}$ first as a narrow diagnostic, at the reference cells only, which is a few minutes of compute rather than a rebuild of the study. Two outcomes are possible, and both are useful:

- **It moves the estimate materially toward $\theta_{\text{true}} = -3.6$ and recovers power.** Then it is a genuine finding, it may weaken the case for Exposure-weighted (since comparable power without committing to a half-life is the more attractive option for a trialist), and it justifies the full four-arm rebuild.
- **It lands on top of Lag-adjusted.** Then one paragraph and a footnote citing the diagnostic close the gap completely, at negligible cost, and the study is defended against the objection without restructuring anything.

Either way the diagnostic is worth running before the decision is taken. Note that there are now two candidate fourth arms, not one: $b_i L_{it}$ as above, and the linear t_{sd} term described in Section 5.3, which the reference implementation already provides and which existing output suggests does recover power. If only one diagnostic is run, the linear t_{sd} term is the better first choice, because it is already implemented, because it is the alternative the original authors offered, and because the `quick-sim` outputs give a prior expectation that it works.

What should not happen is publishing the three-arm comparison with no acknowledgement of the empty cell: as

things stand, a finding that Lag-adjusted fails to recover power is partly a consequence of how the lag device was implemented rather than of the idea itself, and a reader from the crossover tradition is entitled to press on it. If the grid stays at three, the Methods section must state plainly that the lag family is evaluated only in its main-effect-only form.

5.3 A second omission: the original implementation’s own alternative

The reference implementation this program derives from, vendored at `implementations/original/`, supports **three** carryover representations, not two, and none of them is the binary lag indicator. Reading `R/lme_analysis.R`, the fixed-effects formula is assembled at lines 142 to 153 and the exposure variable at lines 108 to 118, giving:

Setting	Fitted exposure terms	Equivalent to
<code>carryover_t1half = 0,</code> <code>simplecarryover = FALSE</code>	$D_{bc} = D_{it}$, so $D_{it} + b_i D_{it}$	Unadjusted (Section 3.1)
<code>carryover_t1half = 0,</code> <code>simplecarryover = TRUE</code>	$D_{it} + b_i D_{it} + \beta_{sd} t_{sd,it}$	no counterpart in this study
<code>carryover_t1half > 0</code>	$D_{bc,it} + b_i D_{bc,it}$, $D_{bc} = (1/2)^{t_{sd}/\hat{t}_{1/2}}$	Exposure-weighted (Section 3.3)

The first and third are mutually exclusive with the second by construction: supplying both a non-zero half-life and `simplecarryover = TRUE` raises an error (lines 67 to 68).

Two consequences follow. First, the binary lag indicator L_{it} evaluated here as Lag-adjusted is *not* inherited from the reference implementation. It was introduced by this project in `simulation-core.R`, following Jones and Kenward. The present study therefore adopts two of the original three specifications and substitutes a device of its own for the third, without evaluating the third itself.

Second, that unevaluated third option, the linear t_{sd} term, is a serious candidate. It sits between the other two on the assumption ladder of Section 4.1, conceding that residual exposure declines with elapsed time, as Exposure-weighted does, while declining to specify a half-life or a decay shape, as Lag-adjusted does. It is also the only alternative the original authors themselves thought worth implementing, and the `quick-sim` outputs discussed in Section 6 give a prior expectation that it recovers power.

A related divergence in the analysis model. Hendrickson’s default `useDE = TRUE` appends the expectancy covariate De whenever expectancy varies across occasions, which it does in the OL+BDC and Hybrid designs though not in CO. The specifications compared here omit it: `simulation-core.R` passes `expectancies` to the design builder, so expectancy shapes the simulated data, but De never enters any of the three fitted formulas. This is orthogonal to the carryover question and applies equally to all three arms, so it should not disturb their ranking, but it does mean the fitted model is not Hendrickson’s, and the close agreement between this study’s 0.860 and Hendrickson’s reported 0.82 at the reference cell has not been checked against that difference.

6. What the choice has already revealed

Recorded because it bears on whether the design succeeded, not as a results summary. Epistemic status: recomputed directly from the stored cell-level summaries under `analysis/data/`, not re-simulated.

How the comparison is scored. The study follows ADEMP, which prescribes a fixed set of performance measures. For a clinical reader the relevant ones are:

- **Power.** How often the trial correctly detects a biomarker effect that is genuinely present. Higher is better; 0.80 is the conventional target.
- **Type I error.** How often it declares an effect when none exists. This should sit at the nominal 0.05, and a method that achieves high power by exceeding 0.05 has not earned it.
- **Bias.** The average gap between the estimate and the true value.
- **Coverage.** How often the 95% confidence interval contains the true value. Should be 0.95.
- **Mean squared error.** A single number combining bias and spread.
- **MCSE.** The simulation’s own margin of error, since each condition is run a finite number of times. Differences smaller than the MCSE should not be interpreted.

Each condition, or **cell**, is one combination of design, sample size, biomarker strength, decay shape and half-life, and is run 500 times.

Within the manuscript’s 216-cell scope, Lag-adjusted and Unadjusted are statistically indistinguishable. Mean power differs by 0.003 against an MCSE of about 0.018, only 3 of 216 conditions differ by more than 0.02, and mean bias (0.373 versus 0.372), mean squared error (1.885 versus 1.876) and confidence-interval coverage (0.936 versus 0.938) agree to three decimal places. The same equality holds in every sensitivity block and in the cluster-robust recalibration cells.

Read through Section 4.2 this is informative rather than merely negative. It says the added-noise mechanism is negligible in this setting, and therefore that the whole of the Exposure-weighted advantage comes from fixing dilution. A two-method comparison could not have reached that conclusion.

One qualification is required, and it turns out to be a naming collision rather than a contradiction. A separate high-precision rerun under `analysis/data/quick-sim/` does show its arm labelled **A3_lagged** ahead of its **A1_binary** arm by 1.3 to 6.5 percentage points on paired tests. That rerun does not fit the specification described in Section 3.2. It calls the package’s `lme_analysis()` with `simplecarryover = TRUE`, which, verified by reading `implementations/original/R/lme_analysis.R:151-153`, appends t_{sd} itself as a linear main effect rather than the binary indicator L_{it} . Lag-adjusted adds $\beta_L L_{it}$; the rerun’s arm adds $\beta_{sd} t_{sd,it}$. A continuous t_{sd} term can track a graded decline across every off-drug occasion; a single binary flag marks one occasion and is constant thereafter. The first carries substantially more information than the second, which is a sufficient explanation for why one gains power and the other does not, and it supersedes the reading that the gap is merely a matter of test calibration. Test size does differ in the same direction, the rerun’s arm carrying the highest false-positive rate of its three, up to 0.053 against 0.037 where 0.05 is nominal, so both effects are probably present. Disentangling them would require fitting both models to the same datasets, which has not been done.

The practical consequence is that the two sources were never measuring the same thing, and the label **A3_lagged** in the `quick-sim` outputs should not be read as this manuscript’s Lag-adjusted specification.

7. Which comparisons this design supports

A choice of methods determines which conclusions the study is entitled to draw. Four rules follow from Sections 3 and 4, and they should govern how the manuscript’s tables are read.

Power and Type I error are comparable across all three. These are properties of the test, not of the coefficient being tested, so they can be compared freely even though the three specifications do not all estimate the same quantity.

Bias, mean squared error and coverage are comparable only within the Unadjusted and Lag-adjusted pair. Those two estimate the coefficient on $b_i D_{it}$; Exposure-weighted estimates the coefficient on $b_i D_{bc,it}$. All three are scored in the manuscript against the single target $\theta_{\text{true}} = -3.6$, which is the correct target for Exposure-weighted but not for the other two. The consequence is easy to misread: Exposure-weighted shows near-zero bias and nominal coverage while the other two show substantial bias and under-coverage, and that gap is largely a statement about which quantity each one targets rather than about which estimator is better behaved. Those columns should not be used to rank the three.

The Lag-adjusted versus Unadjusted contrast is the controlled one. Same estimand, same simulated patients, one term of difference. Whatever separates them is the added-noise mechanism, cleanly identified. This is the single most trustworthy comparison in the study, and it is the reason the third arm was worth adding.

Any conclusion about the lag device is bounded by its main-effect form. Because the specification lacks $b_i L_{it}$, a finding that Lag-adjusted does not recover power says that a lag *main effect* does not recover power. It does not establish that the crossover tradition’s approach fails, only that this implementation of it does. Section 5.2 sets out what would be needed to close that gap.

8. Summary

The three specifications are Unadjusted, which codes exposure as a simple on/off switch; Lag-adjusted, which adds a flag for the first off-drug measurement; and Exposure-weighted, which replaces the switch with a dimmer that fades at an assumed half-life. They are the right three because they occupy the three distinct positions

available on the unknown washout curve, because two of them estimate the same quantity and so isolate one damage mechanism under otherwise identical conditions while the third addresses the other, and because each is the working default of a literature that will read the result. Their principal weakness is the empty cell at $b_i L_{it}$, which leaves the crossover tradition's remedy represented only in a form that cannot address dilution, and which should either be filled or explicitly disclaimed.

9. Proposed additions to NOTATION.md

Three gaps in the canonical notation are exposed by this comparison.

9.1 Add L_{it} . The lagged just-off-drug indicator has no symbol in Part 1. Proposed entry for the treatment-exposure table:

Symbol	Meaning
L_{it}	lagged just-off-drug indicator, $D_{i,t-1}(1 - D_{it})$; 1 at the first off-drug occasion following an on-drug one (code column L)

9.2 Split the estimand symbol. Part 1 currently carries a single entry, $\beta_{bm:D}$, glossed as the biomarker-by-treatment interaction coefficient and annotated **bm:Dbc**. That conflates two different coefficients. Unadjusted and Lag-adjusted estimate the coefficient on $b_i D_{it}$; Exposure-weighted estimates the coefficient on $b_i D_{bc,it}$. The distinction is what licenses the controlled comparison of Section 4.2, and it is also why the manuscript's bias, mean-squared-error and coverage columns are comparable within the first pair but not across to the third. Proposed replacement:

Symbol	Meaning
$\beta_{bm:D}$	interaction coefficient on the binary regressor, $b_i D_{it}$ (bm:Db); the estimand of Unadjusted and Lag-adjusted
$\beta_{bm:D_{bc}}$	interaction coefficient on the exposure-decayed regressor, $b_i D_{bc,it}$ (bm:Dbc); the estimand of Exposure-weighted

9.3 Add the main-effect coefficients. Part 1 has no symbols for the intercept or for the non-interaction terms of the analysis model, so any paper writing the model out must invent them. Proposed: β_0 intercept, β_b biomarker main effect, β_t time, β_D drug-exposure main effect, and β_L lagged-indicator main effect. The entry for β_b should note explicitly that it is distinct from β_{bm} , which the file reserves for the moderation parameter of the mean-moderation architecture; the two are easily confused and mean quite different things.

9.4 A policy tension to resolve before renaming. Part 2 of NOTATION.md requires that a stored data value doubling as a reporting label be kept identical to it, and records **spec** as stored **E1/E2/E3** and reported unmapped. That rule was enforced on 2026-07-31, when twenty-one **.rds** files were migrated from **A1/A2/A3** precisely to retire a display-time mapping. Renaming the specifications reintroduces the layer that migration removed.

The same file supplies the governing precedent, however: the architectures were renamed to mechanism names while their stored values (**mean_moderation**, **mvn**, **combined**) were left untouched and the mapping recorded. That is the operation proposed here, for the same reason. Follow that precedent rather than migrating twice. Keep **E1/E2/E3** stored permanently, apply display names through one shared helper, and mark the Part 2 row as mapped. A second migration would invalidate the archived drafts, the pre-migration backups, and every summary quoted in papers 01 and 10, all to satisfy a rule the file already suspends for the architectures.

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